

Name: _____

These questions assess your understanding of the material from Chapters 13, 14, 15 of Silberberg. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must explain why you chose your answer. Remember to report the units and correct number of significant figures in your answers.

1. Calculate the work output of a Carnot engine operating between temperatures of 582 K and 302 K for 7250 J of heat transfer to the engine.

ANSWER: 3488 J ✓

$$\eta_c = 1 - \frac{T_c}{T_h} = \frac{W}{Q_h}$$

$$W = Q_h \left(1 - \frac{T_c}{T_h}\right) = 7250 \text{ J} \left(1 - \frac{302 \text{ K}}{582 \text{ K}}\right) = 3487.97 \text{ J} = 3488 \text{ J}$$

2. Aaron's 62-L (16.4-gal) steel gasoline tank is full of gas. Both the tank and the gasoline have an initial temperature of 1.6°C. How much gasoline has leaked out due to thermal expansion by the time both the tank and the gasoline have warmed to 30.9°C? The coefficient of volume expansion for steel is $35 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ and for gasoline it is $950 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$. Show all work.

ANSWER: 1.66 L ✓

$$\Delta V_L = \text{Volume leaked out}$$

$$\Delta V_L = \Delta V_g - \Delta V_s = \beta_g V_0 \Delta T - \beta_s V_0 \Delta T = V_0 \Delta T (\beta_g - \beta_s)$$

$$\Delta V_L = (62 \text{ L})(29.3^\circ\text{C}) \left(950 \times 10^{-6} \frac{1}{^\circ\text{C}} - 35 \times 10^{-6} \frac{1}{^\circ\text{C}}\right) = 1.66 \text{ L}$$

$$\Delta T = 30.9^\circ\text{C} - 1.6^\circ\text{C} = 29.3^\circ\text{C}$$

3. Find the increase in entropy as 5.5 kg of ice, originally at 0.9°C, melts to form water at 0.9°C at a particular pressure. The latent heat of fusion for water is 334 kJ/kg.

ANSWER: 6703.2 J/K ✓

$$\Delta S = \frac{Q}{T} = \frac{mL_f}{T}$$

$$\Delta S = \frac{(5.5 \text{ kg})(334 \times 10^3 \text{ J/kg})}{274.05 \text{ K}}$$

$$\Delta S = 6703.16 \text{ J/K}$$

$$\frac{334 \text{ kJ} / 1000 \text{ J}}{\text{kg} / 1 \text{ kg}} = 334 \times 10^3 \text{ J/kg}$$

$$0.9 + 273.15 = 274.05 \text{ K}$$

+3

4. Nick wants to use his pressure cooker to cook food in liquid water at a temperature of 130°C . To determine the pressure required inside the pressure cooker for water to boil at this temperature, he can use the Clausius-Clapeyron equation for water: $P(T) = (1 \text{ kPa}) \cdot \exp\left[(4750 \text{ K})\left(\frac{1}{273.15 \text{ K}} - \frac{1}{T}\right)\right]$. In this equation, T is the temperature in Kelvin and P is the pressure in kPa. Given that the lid of the pressure cooker is a disk with a diameter of 36 cm, what is the force that the lid must be able to withstand at the required pressure?

$$P = \frac{F}{A} \rightarrow F = P \cdot A$$

ANSWER: 17.42 kN

$$P(T) = P_g + P_{\text{atm}} \rightarrow P_g = P(T) - P_{\text{atm}}$$

$$130 + 273.15 = 403.15$$

$$36 \text{ cm} = 0.36 \text{ m}$$

$$P_g = (1 \text{ kPa}) \cdot e^{(4750) \left(\frac{1}{273.15} - \frac{1}{403.15} \right)} - 101.325 \text{ kPa}$$

$$P_g = 171.13656 \text{ kPa}$$

$$A = \pi r^2 = \pi \left(\frac{d}{2} \right)^2 = \pi \left(\frac{0.36 \text{ m}}{2} \right)^2 = 0.101788 \text{ m}^2$$

$$F = (171.13656 \text{ kPa})(0.101788 \text{ m}^2) = 17.42 \text{ kN}$$

5. Dylan is producing heat at a rate of about 299 W while doing light exercise. At what rate must water evaporate from his body to get rid of the produced energy? The latent heat of vaporization of water is 2430 kJ/kg at body temperature, 37°C .

ANSWER: 7.38 g/min

$$Q = mL_v \rightarrow \frac{Q}{t} = \frac{mL_v}{t} = P$$

$$P = \dot{m}L_v \rightarrow \dot{m} = \frac{P}{L_v} = \frac{299 \frac{\text{J}}{\text{s}}}{2430 \times 10^3 \frac{\text{J}}{\text{kg}}} = 1.230 \times 10^{-4} \frac{\text{kg}}{\text{s}}$$

$$\frac{1.230 \times 10^{-4} \frac{\text{kg}}{\text{s}}}{1 \text{ kg}} \left| \frac{1000 \text{ g}}{1 \text{ kg}} \right| \left| \frac{60 \text{ s}}{1 \text{ min}} \right| = 7.38 \text{ g/min}$$

6. Suppose that there is heat transfer of 49 J to a system, while the system does 60 J of work. Later, there is heat transfer of 40 J out of the system while 30 J of work is done on the system. What is the net change in internal energy of the system?

ANSWER: -21 J

$$\Delta U = Q + W$$

$$Q_1 = 49 \text{ J} \quad Q_2 = -40 \text{ J}$$

$$W_1 = -60 \text{ J} \quad W_2 = 30 \text{ J}$$

$$\Delta U = (49 - 40) \text{ J} + (-60 \text{ J} + 30 \text{ J}) = 9 \text{ J} - 30 \text{ J} = -21 \text{ J}$$

7. Efe is a design engineer for a lightbulb company. He manufactures a gas-filled incandescent light bulb so that the gas inside the bulb is at atmospheric pressure when the bulb has a temperature of 8°C . What is the pressure in the light bulb when it rises to 282°C ? The coefficient of volume expansion for the glass light bulb is $32 \times 10^{-6} \text{ } 1^\circ\text{C}$.

ANSWER: 198.33 kPa ✓

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\Delta V = \beta V_1 \Delta T \rightarrow V_2 - V_1 = \beta V_1 \Delta T \rightarrow V_2 = \beta V_1 \Delta T + V_1$$

$$V_2 = V_1 (\beta \Delta T + 1)$$

$$282 + 273.15 = 555.15 \text{ K}$$

$$8 + 273.15 = 281.15 \text{ K}$$

$$282 - 8 = 274^\circ\text{C}$$

$$P_2 = \frac{P_1 V_1 T_2}{T_1 V_2}$$

$$\rightarrow P_2 = \frac{P_1 T_2}{T_1 (\beta \Delta T + 1)} = \frac{P_1 T_2}{T_1 (\beta \Delta T + 1)}$$

$$P_2 = \frac{(101325 \text{ Pa})(555.15 \text{ K})}{(281.15 \text{ K})(32 \times 10^{-6} \frac{1}{^\circ\text{C}} \cdot 274^\circ\text{C} + 1)} = 198.33 \text{ kPa}$$

8. In a particular year, the main span of San Francisco's Golden Gate Bridge was 1276-m long on the coldest day. The bridge was exposed to temperatures ranging from -3°C to 35°C . What is the change in its length between these temperatures? Assume that the bridge is made entirely of steel so that its coefficient of linear expansion is $12 \times 10^{-6} \text{ } 1^\circ\text{C}$.

ANSWER: 0.582 m ✓

$$\Delta L = \alpha L \Delta T$$

$$\Delta T = 35^\circ\text{C} - (-3^\circ\text{C}) = 38^\circ\text{C}$$

$$\Delta L = (12 \times 10^{-6} \frac{1}{^\circ\text{C}})(1276 \text{ m})(38^\circ\text{C}) = 0.582 \text{ m}$$

9. Christopher puts a 0.77-kg aluminum pot on the stove and fills it with 0.35 liters of water. Both the pot and the water are initially at 1.4°C . How much heat is required to bring them both to 76.5°C ? The specific heat value of water is $4186 \text{ J/kg}^\circ\text{C}$ and for aluminum it is $900 \text{ J/kg}^\circ\text{C}$.

ANSWER: 162.1 kJ ✓

$$Q_{\text{required}} = Q_A + Q_W = m_A c_A \Delta T + m_W c_W \Delta T$$

$$\Delta T = 76.5^\circ\text{C} - 1.4^\circ\text{C} = 75.1^\circ\text{C}$$

$$0.35 \text{ L} \left| \frac{1 \text{ m}^3}{1000 \text{ L}} \right| = 0.35 \times 10^{-3} \text{ m}^3 \quad m = \rho V_W$$

$$0.35 \times 10^{-3} \cdot 1000 \frac{\text{kg}}{\text{m}^3} = 0.35 \text{ kg}$$

$$(0.77 \text{ kg} \cdot 900 \frac{\text{J}}{\text{kg}^\circ\text{C}} \cdot 75.1^\circ\text{C}) + (0.35 \text{ kg} \cdot 4186 \frac{\text{J}}{\text{kg}^\circ\text{C}} \cdot 75.1^\circ\text{C})$$

$$Q_{\text{required}} = 162.1 \text{ kJ}$$

+3

Answers for exam 6.capitalize() (Chapters: 13, 14, 15)

1. 3487.97 J
2. 1.66 L
3. 6.7 kJ/K
4. 17.42 kN
5. 7.38 g/min
6. -21 J
7. 198.334 kPa
8. 58.19 cm
9. 162.07 kJ